

[Paper 3 continued from p. 15.]

we find

$$\Sigma \left(\frac{d^2 \xi}{da^2} \right) = \frac{2\Sigma \frac{1}{\xi+h^2}}{\Sigma \frac{a^2}{(\xi+h^2)^2}},$$

and we hence obtain

$$\nabla \chi \xi = \frac{2\chi' \xi \Sigma \frac{1}{\xi+h^2} + 4\chi'' \xi}{\Sigma \frac{a^2}{(\xi+h^2)^2}}.$$

Hence the function

$$\int_0^\infty d\phi \cdot \nabla \frac{1}{(\xi + \phi) \sqrt{\{(\xi + h^2 + \phi) \dots\}}}$$

(observing that differentiation with respect to ξ is the same as differentiation with respect to ϕ) becomes integrable, and taking the integral between the proper limits, its value is

$$- \frac{2\chi'_0 \xi \Sigma \frac{1}{\xi+h^2} + 4\chi''_0 \xi}{\Sigma \frac{a^2}{(\xi+h^2)^2}},$$

where

$$\chi_0 \xi = \frac{1}{\xi \sqrt{\{(\xi + h^2)(\xi + h_1^2) \dots\}}}.$$

We have immediately

$$\frac{\chi'_0 \xi}{\chi_0 \xi} = -\frac{1}{2} \left(\frac{2}{\xi} + \Sigma \frac{1}{\xi + h^2} \right);$$

or

$$2\chi_0 \xi \Sigma \left(\frac{1}{\xi + h^2} \right) + 4\chi'_0 \xi = -4 \frac{\chi_0 \xi}{\xi};$$

whence

$$\int_0^\infty d\phi \cdot \nabla \frac{1}{(\xi + \phi) \sqrt{\{(\xi + h^2 + \phi) \dots\}}} = \frac{4}{\xi^2 \sqrt{\{(\xi + h^2)(\xi + h_1^2) \dots\}}} \left\{ \frac{a^2}{(\xi + h^2)^2} + \frac{b^2}{(\xi + h_1^2)^2} + \dots \right\}.$$

Hence restoring the value of ∇ , and of the first side of the equation,

$$\begin{aligned} & \iint \dots (\text{n times}) \frac{dx \, dy \dots}{\{(a-x)^2 + (b-y)^2 \dots\}^{\frac{1}{2}\text{n}+q}} \\ &= \frac{hh_1 \dots \pi^{\frac{1}{2}\text{n}}}{2^{2q-2} \cdot 1.2 \dots q \cdot \Gamma(\frac{1}{2}\text{n} + q)} \left(\frac{d^2}{da^2} + \frac{d^2}{db^2} + \dots \right)^{q-1} \\ & \quad \times \frac{1}{\xi^2 \sqrt{\{(\xi + h^2)(\xi + h_1^2) \dots\}}} \left\{ \frac{a^2}{(\xi + h^2)^2} + \frac{b^2}{(\xi + h_1^2)^2} + \dots \right\}, \end{aligned}$$

with the condition

$$\frac{a^2}{\xi + h^2} + \frac{b^2}{\xi + h_1^2} + \dots = 1;$$

from which equation the differential coefficients of ξ , which enter into the preceding result, are to be determined.

In general if u be any function of $\xi, a, b \dots$

$$\left(\frac{d^2}{da^2} + \frac{d^2}{db^2} + \dots \right) u = \frac{4 \frac{d^2 u}{d\xi^2} + 2 \frac{du}{d\xi} \Sigma \frac{1}{\xi + h^2} + 4 \frac{d^2 u}{d\xi da} \Sigma \frac{a}{\xi + h^2}}{\Sigma \frac{a^2}{(\xi + h^2)^2}} + \Sigma \frac{d^2 u}{da^2},$$

from which the values of the second side for $q = 1, q = 2, \&c.$ may be successively calculated.

The performance of the operation $\left(\frac{d}{da} \right)^p \left(\frac{d}{db} \right)^q \left(\frac{d}{dc} \right)^r \dots$, upon the integral V' , leads in like manner to a very great number of integrals, all of them expressible algebraically, for a single differentiation renders the integration with respect to ϕ possible. But this is a subject which need not be further considered at present.

We shall consider, lastly, the definite integral

$$U = \iint \dots (\mathfrak{n} \text{ times}) \frac{(a-x)f\left(\frac{x^2}{h^2} + \frac{y^2}{h_1^2} + \dots\right) dx dy \dots}{\{(a-x)^2 + (b-y)^2 \dots\}^{\frac{1}{2}\mathfrak{n}}},$$

limits, &c. as before. This is readily deduced from the less general one

$$\iint \dots (\mathfrak{n} \text{ times}) \frac{(a-x) dx dy \dots}{\{(a-x)^2 + (b-y)^2 \dots\}^{\frac{1}{2}\mathfrak{n}}}.$$

For representing this quantity by $F(h, h_1, \dots)$, it may be seen that

$$U = \int_0^1 f(m^2) \frac{d}{dm} F(mh, mh_1, \dots) dm;$$

but in the value of $F(h, h_1, \dots)$, changing $h, h_1, \dots$ into $mh, mh_1, \dots$ also writing $m^2\phi$ instead of ϕ , and $m^2\xi'$ for ξ , we have

$$F(mh, mh_1, \dots) = \frac{hh_1 \dots \pi^{\frac{1}{2}\mathfrak{n}}}{\Gamma(\frac{1}{2}\mathfrak{n})} a \int_0^\infty \frac{d\phi}{(\xi' + h^2 + \phi)\sqrt{\Phi'}},$$

where

$$\Phi' = (\xi' + h^2 + \phi)(\xi' + h_1^2 + \phi) \dots$$

and

$$\frac{a^2}{\xi' + h^2} + \frac{b^2}{\xi' + h_1^2} + \dots = m^2.$$

Hence

$$\frac{d}{dm} F(mh, mh_1, \dots) = \frac{hh_1 \dots \pi^{\frac{1}{2}\mathfrak{n}}}{\Gamma(\frac{1}{2}\mathfrak{n})} a \frac{d\xi'}{dm} \int_0^\infty d\phi \frac{d}{d\xi'} \frac{1}{(\xi' + h^2 + \phi)\sqrt{\Phi'}},$$

or, observing that $\frac{d}{d\xi'}$ is equivalent to $\frac{d}{d\phi}$, and effecting the integration between the proper limits,

$$\frac{d}{dm} F(mh, mh_1, \dots) = - \frac{hh_1 \dots \pi^{\frac{1}{2}\mathfrak{n}}}{\Gamma(\frac{1}{2}\mathfrak{n})} a \frac{d\xi'}{dm} \frac{1}{(\xi' + h^2) \sqrt{\{(\xi' + h^2)(\xi' + h_1^2) \dots\}}}.$$

Substituting this value, also $f \left\{ \frac{a^2}{\xi' + h^2} + \frac{b^2}{\xi' + h_1^2} + \dots \right\}$ for $f(m^2)$, in the value of U , and observing that $m = 0$ gives $\xi' = \infty$, $m = 1$ gives $\xi' = \xi$, where ξ is a quantity determined as before by the equation

$$\frac{a^2}{\xi + h^2} + \frac{b^2}{\xi + h_1^2} + \dots = 1,$$

we have

$$U = - \frac{hh_1 \dots \pi^{\frac{1}{2}\mathbf{n}}}{\Gamma(\frac{1}{2}\mathbf{n})} a \int_{\infty}^{\xi} \frac{f \left\{ \frac{a^2}{\xi' + h^2} + \dots \right\} d\xi'}{(\xi' + h^2) \sqrt{\{(\xi' + h^2)(\xi' + h_1^2) \dots\}}};$$

or writing $\phi + \xi$ for ξ' , $d\xi' = d\phi$, the limits of ϕ are $0, \infty$; or, inverting the limits and omitting the negative sign,

$$U = \frac{hh_1 \dots \pi^{\frac{1}{2}\mathbf{n}}}{\Gamma(\frac{1}{2}\mathbf{n})} a \int_0^{\infty} \frac{f \left\{ \frac{a^2}{\xi + h^2 + \phi} + \frac{b^2}{\xi + h_1^2 + \phi} + \dots \right\} d\phi}{(\xi + h^2 + \phi) \sqrt{\{(\xi + h^2 + \phi)(\xi + h_1^2 + \phi) \dots\}}};$$

which, in the particular case of $\mathbf{n} = 3$, may easily be made to coincide with known results. The analogous integral

$$\iint \dots (\mathbf{n} \text{ times}) \frac{f \left(\frac{x^2}{h^2} + \frac{y^2}{h_1^2} + \dots \right) dx dy \dots}{\{(a - x)^2 + (b - y)^2 \dots\}^{\frac{1}{2}\mathbf{n}}}$$

is apparently not reducible to a single integral.

4.

ON CERTAIN EXPANSIONS, IN SERIES OF MULTIPLE SINES
AND COSINES.

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IN the following paper we shall suppose e the base of the hyperbolic system of logarithms; ϵ a constant, such that its modulus, and also the modulus of $\frac{1}{e}\{1 - \sqrt{1 - e^2}\}$, are each of them less than unity; $\chi\{e^{u\sqrt{-1}}\}$ a function of u , which, as u increases from 0 to π , passes continuously from the former of these values to the latter, without becoming a maximum in the interval, $f\{e^{u\sqrt{-1}}\}$ any function of u which remains finite and continuous for values of u included between the above limits. Hence, writing

$$\chi\{e^{u\sqrt{-1}}\} = m, \quad (1)$$

and considering the quantity

$$\frac{\sqrt{1 - e^2} f\{e^{u\sqrt{-1}}\}}{\sqrt{-1} e^{u\sqrt{-1}} \chi'\{e^{u\sqrt{-1}}\} (1 - e \cos u)} \quad (2)$$

as a function of m , for values of m or u included between the limits 0 and π , we have

$$\frac{\sqrt{1 - e^2} f\{e^{u\sqrt{-1}}\}}{\sqrt{-1} e^{u\sqrt{-1}} \chi'\{e^{u\sqrt{-1}}\} (1 - e \cos u)} = \frac{2}{\pi} \sum_{-\infty}^{\infty} \cos rm \int_0^{\pi} \frac{\sqrt{1 - e^2} f\{e^{u\sqrt{-1}}\} \cos rm \, dm}{\sqrt{-1} e^{u\sqrt{-1}} \chi'\{e^{u\sqrt{-1}}\} (1 - e \cos u)} \quad (3)$$

(Poisson, *Méc.* tom. I. p. 650); which may also be written

$$\frac{\sqrt{1 - e^2} f\{e^{u\sqrt{-1}}\}}{\sqrt{-1} e^{u\sqrt{-1}} \chi'\{e^{u\sqrt{-1}}\} (1 - e \cos u)} = \frac{2}{\pi} \sum_{-\infty}^{\infty} \cos rm \int_0^{\pi} \frac{\sqrt{1 - e^2} f\{e^{u\sqrt{-1}}\} \cos r \chi\{e^{u\sqrt{-1}}\} \, du}{1 - e \cos u}; \quad (4)$$

and if the first side of the equation be generally expansible in a series of multiple cosines of m , instead of being so in particular cases only, its expanded value will always be the one given by the second side of the preceding equation.

Now, between the limits 0 and π , the function

$$f\{e^{u\sqrt{-1}}\} \cos r\chi\{e^{u\sqrt{-1}}\}$$

will always be expansible in a series of multiple cosines of u ; and if by any algebraical process the function $f\rho \cos r\chi\rho$ can be expanded in the form

$$f\rho \cos r\chi\rho = \Sigma_{-\infty}^{\infty} a_s \rho^s, \quad (a_s = a_{-s}); \quad (5)$$

we have, in a convergent series,

$$f\{e^{u\sqrt{-1}}\} \cos r\chi\{e^{u\sqrt{-1}}\} = a_0 + 2\Sigma_1^{\infty} a_s \cos su. \quad (6)$$

Again, putting

$$\frac{1}{e}\{1 - \sqrt{1 - e^2}\} = \lambda, \quad (7)$$

we have

$$\frac{\sqrt{1 - e^2}}{1 - e \cos u} = 1 + 2\Sigma_1^{\infty} \lambda^p \cos pu. \quad (8)$$

Multiplying these two series, and effecting the integration, we obtain

$$\frac{1}{\pi} \int_0^{\pi} \frac{\sqrt{1 - e^2} f\{e^{u\sqrt{-1}}\} \cos r\chi\{e^{u\sqrt{-1}}\} du}{1 - e \cos u} = 2 \left\{ \frac{1}{2} a_0 + \Sigma_1^{\infty} (a_s \lambda^s) \right\}. \quad (9)$$

And the second side of this equation being obviously derived from the expansion of $f\lambda \cos r\chi\lambda$ by rejecting negative powers of λ and dividing by 2, the term independent of λ may conveniently be represented by the notation

$$\overline{2f\lambda \cos r\chi\lambda}; \quad (10)$$

where in general, if $\Gamma\lambda$ can be expanded in the form

$$\Gamma\lambda = \Sigma_{-\infty}^{\infty} (A_s \lambda^s), \quad [A_{-s} = A_s], \quad (11)$$

we have

$$\overline{\Gamma\lambda} = \frac{1}{2} A_0 + \Sigma_1^{\infty} A_s \lambda^s. \quad (12)$$

(By what has preceded, the expansion of $\Gamma\lambda$ in the above form is always possible in a certain sense; however, in the remainder of the present paper, $\Gamma\lambda$ will always be of a form to satisfy the equation $\Gamma(\frac{1}{\lambda}) = \Gamma\lambda$, except in cases which will afterwards be considered, where the condition $A_{-s} = A_s$ is unnecessary.)

Hence, observing the equations (4), (9), (10),

$$\frac{\sqrt{1 - e^2} f\{e^{u\sqrt{-1}}\}}{\sqrt{-1} e^{u\sqrt{-1}} \chi'\{e^{u\sqrt{-1}}\} (1 - e \cos u)} = \Sigma_{-\infty}^{\infty} \cos rm \overline{2 \cos r\chi\lambda f\lambda}; \quad (13)$$

from which, assuming a system of equations analogous to (1), and representing by $\Pi(\Phi)$ the product $\Phi_1 \Phi_2 \dots$, it is easy to deduce

$$\begin{aligned} & \Pi \left\{ \frac{\sqrt{1 - e^2}}{\sqrt{-1} e^{u\sqrt{-1}} \chi'\{e^{u\sqrt{-1}}\} (1 - e \cos u)} \right\} f\{e^{u_1\sqrt{-1}}, e^{u_2\sqrt{-1}} \dots\} \\ & = \Sigma_{-\infty}^{\infty} \Sigma_{-\infty}^{\infty} \dots \Pi \cos rm \overline{\Pi(2 \cos r\chi\lambda) f(\lambda_1, \lambda_2 \dots)}. \end{aligned} \quad (14)$$

where $\Gamma(\lambda_1, \lambda_2 \dots)$ being expansible in the form

$$\Gamma(\lambda_1, \lambda_2 \dots) = \Sigma_{-\infty}^{\infty} \Sigma_{-\infty}^{\infty} \dots A_{s_1, s_2, \dots} \lambda_1^{s_1} \lambda_2^{s_2} \dots, \quad [A_{s_1, s_2, \dots} = A_{-s_1, -s_2, \dots}], \quad (15)$$

$$\overline{\Gamma(\lambda_1, \lambda_2 \dots)} = \Sigma_0^{\infty} \Sigma_0^{\infty} \dots \frac{1}{2^N} A_{s_1, s_2, \dots} \lambda_1^{s_1} \lambda_2^{s_2} \dots, \quad (16)$$

N being the number of exponents which vanish.

The equations (13) and (14) may also be written in the forms

$$f\{e^{u\sqrt{-1}}\} = \Sigma_{-\infty}^{\infty} \cos rm \overline{2 \cos r\chi\lambda \frac{\sqrt{-1}\chi'\lambda\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}}{\sqrt{1 - e^2}}} f\lambda, \quad (17)$$

$$f\{e^{u_1\sqrt{-1}}, e^{u_2\sqrt{-1}} \dots\} = \Sigma_{-\infty}^{\infty} \Sigma_{-\infty}^{\infty} \dots \Pi(\cos rm) \overline{\Pi\left\{2 \cos r\chi\lambda \frac{\sqrt{-1}\chi'\lambda\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}}{\sqrt{1 - e^2}}\right\}} f(\lambda_1, \lambda_2 \dots). \quad (18)$$

As examples of these formulae, we may assume

$$\chi\{e^{u\sqrt{-1}}\} = m = u - e \sin u. \quad (19)$$

Hence, putting

$$\lambda^r e^{-\frac{re}{2}(\lambda - \lambda^{-1})} \lambda^{-r} e^{\frac{re}{2}(\lambda - \lambda^{-1})} = A_r, \quad (20)$$

and observing the equation

$$\sqrt{-1} e^{u\sqrt{-1}} \chi'\{e^{u\sqrt{-1}}\} = 1 - e \cos u, \quad (21)$$

the equation (17) becomes

$$f\{e^{u\sqrt{-1}}\} = \Sigma_{-\infty}^{\infty} \cos rm \overline{A_r \frac{\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}^2}{\sqrt{(1 - e^2)}}} f\lambda. \quad (22)$$

Thus, if

$$\theta - \varpi = \cos^{-1} \frac{\cos u - e}{1 - e \cos u}, \quad (23)$$

assuming

$$f\{e^{u\sqrt{-1}}\} = \frac{\cos u - e}{1 - e \cos u}, \quad (24)$$

$$\cos(\theta - \varpi) = \Sigma_{-\infty}^{\infty} \frac{1}{\sqrt{1 - e^2}} \cos rm \overline{\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\} \{\frac{1}{2}(\lambda + \lambda^{-1}) - e\} A_r}. \quad (25)$$

the term corresponding to $r = 0$ being

$$\frac{1}{2\sqrt{1 - e^2}} \{2\lambda - 2e - e(\lambda^2 + 1) + 2e\lambda\} = -e. \quad (26)$$

Again, assuming

$$f\{e^{u\sqrt{-1}}\} = \frac{d\theta}{dm} = \frac{\sqrt{1 - e^2}}{(1 - e \cos u)^2}, \quad (27)$$

and integrating the resulting equation with respect to m ,

$$\theta - \varpi = \Sigma_{-\infty}^{\infty} \frac{\sin rm}{r} \overline{A_r} = m + 2\Sigma_1^{\infty} \frac{\sin rm}{r} \overline{A_r}, \quad (28)$$

a formula given in the fifth No. of the *Mathematical Journal*, and which suggested the present paper.

As another example, let

$$f\{e^{u\sqrt{-1}}\} = \cos(\theta - \varpi) \frac{d\theta}{dm} = \frac{\sqrt{1-e^2}(\cos u - e)}{(1 - e \cos u)^3}. \quad (29)$$

Then integrating with respect to m , there is a term

$$2m \frac{\overline{\frac{1}{2}(\lambda + \lambda^{-1}) - e}}{1 - \frac{1}{2}e(\lambda + \lambda^{-1})}, \quad (30)$$

which it is evident, *à priori*, must vanish. Equating it to zero, and reducing, we obtain

$$\frac{e}{1 - e^2} = \frac{\overline{\lambda + \lambda^{-1}}}{1 - \frac{1}{2}e(\lambda + \lambda^{-1})}, \quad (31)$$

that is

$$\frac{e}{1 - e^2} = \lambda + \frac{e}{2}(\lambda^2 + 1) + \frac{e^2}{4}(\lambda^3 + 3\lambda) + \frac{e^3}{8}(\lambda^4 + 4\lambda^2 + 3) + \dots, \quad (32)$$

a singular formula, which may be verified by substituting for λ its value: we then obtain

$$\sin(\theta - \varpi) = 2\Sigma_1^{\infty} \frac{\sin rm}{r} \overline{A_r \frac{\frac{1}{2}(\lambda + \lambda^{-1}) - e}{1 - \frac{1}{2}e(\lambda + \lambda^{-1})}}. \quad (33)$$

The expansions of $\sin k(\theta - \varpi)$, $\cos k(\theta - \varpi)$, are in like manner given by the formulae

$$\cos k(\theta - \varpi) = \Sigma_{-\infty}^{\infty} \overline{A_r L' \cos kL} \cos rm, \quad (34)$$

$$\sin k(\theta - \varpi) = \Sigma_{-\infty}^{\infty} \overline{A_r \frac{1}{kr} \cos kL} \frac{\sin rm}{r}, \quad (35)$$

where, to abbreviate, we have written

$$\cos^{-1} \left\{ \frac{\frac{1}{2}(\lambda + \lambda^{-1}) - e}{1 - \frac{1}{2}e(\lambda + \lambda^{-1})} \right\} = L, \quad (36)$$

$$\frac{\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}^2}{\sqrt{1 - e^2}} = L'. \quad (37)$$

Forming the analogous expressions for

$$\cos k(\theta' - \varpi'), \quad \sin k(\theta' - \varpi'),$$

substituting in

$$\begin{aligned} \cos k(\theta - \theta') = & \cos k(\varpi - \varpi') \{ \cos k(\theta - \varpi) \cos k(\theta' - \varpi') + \sin k(\theta - \varpi) \sin k(\theta' - \varpi') \} \\ & - \sin k(\varpi - \varpi') \{ \sin k(\theta - \varpi) \cos k(\theta' - \varpi') - \sin k(\theta' - \varpi') \cos k(\theta - \varpi) \}, \end{aligned}$$

and reducing the whole to multiple cosines, the final result takes the very simple form

$$\cos k(\theta - \theta') = \Sigma_{-\infty}^{\infty} \cos \{ r' m' - r m + k(\varpi - \varpi') \} \overline{A_r A_{r'} \cos kL \cos kL' \left(L - \frac{1}{kr} \right) \left(L' - \frac{1}{kr'} \right)}. \quad (38)$$

Again, formulae analogous to (14), (18), may be deduced from the equation

$$\begin{aligned} \Gamma(m_1, m_2 \dots) = & \Sigma_{-\infty}^{\infty} \Sigma_{-\infty}^{\infty} \dots \left\{ \cos(r_1 m_1 + r_2 m_2 \dots) \int_0^{2\pi} \frac{dm_1}{2\pi} \int_0^{2\pi} \frac{dm_2}{2\pi} \dots \right. \\ & \times \cos(r_1 m_1 + r_2 m_2 \dots) \Gamma(m_1, m_2 \dots) \\ & + \sin(r_1 m_1 + r_2 m_2 \dots) \int_0^{2\pi} \frac{dm_1}{2\pi} \int_0^{2\pi} \frac{dm_2}{2\pi} \dots \\ & \left. \times \sin(r_1 m_1 + r_2 m_2 \dots) \Gamma(m_1, m_2 \dots) \right\}, \end{aligned} \quad (39)$$

which holds from $m_i = 0$ to $m_i = 2\pi$, &c., but in many cases universally. In this case, writing for $\Gamma(m_1, m_2 \dots)$ the function

$$\Pi \left\{ \frac{1}{\sqrt{-1} e^{u\sqrt{-1}} \chi' \{ e^{u\sqrt{-1}} \}} \frac{\sqrt{1-e^2} - e \sin u \sqrt{-1}}{1 - e \cos u} \right\} f \{ e^{u_1 \sqrt{-1}}, e^{u_2 \sqrt{-1}} \dots \}; \quad (40)$$

and observing

$$\frac{\sqrt{1-e^2} - e \sin u \sqrt{-1}}{1 - e \cos u} = \frac{1 + \lambda e^{-u\sqrt{-1}}}{1 - \lambda e^{-u\sqrt{-1}}} = 1 + 2 \Sigma_1^{\infty} \{ \cos su - \sqrt{-1} \sin su \} \lambda^s, \quad (41)$$

an exactly similar analysis, (except that in the expansion

$$\Gamma(\lambda_1, \lambda_2 \dots) = \Sigma_{-\infty}^{\infty} \Sigma_{-\infty}^{\infty} \dots A_{s_1, s_2, \dots} \lambda_1^{s_1} \lambda_2^{s_2} \dots,$$

the supposition is not made that $A_{s_1, s_2, \dots} = A_{-s_1, -s_2, \dots}$), leads to the result

$$\begin{aligned} & f \{ e^{u_1 \sqrt{-1}}, e^{u_2 \sqrt{-1}} \dots \} \Pi \left\{ \frac{\sqrt{1-e^2} - e \sin u \sqrt{-1}}{\sqrt{-1} e^{u\sqrt{-1}} \chi' \{ e^{u\sqrt{-1}} \}} (1 - e \cos u) \right\} \\ & = \Sigma_{-\infty}^{\infty} \Sigma_{-\infty}^{\infty} \dots \left\{ \cos(r_1 m_1 + r_2 m_2 \dots) \overline{2^n \cos(r_1 \chi \lambda_1 + r_2 \chi \lambda_2 \dots) f(\lambda_1, \lambda_2 \dots)} \right. \\ & \quad \left. + \sin(r_1 m_1 + r_2 m_2 \dots) \overline{2^n \sin(r_1 \chi \lambda_1 + r_2 \chi \lambda_2 \dots) f(\lambda_1, \lambda_2 \dots)} \right\}, \end{aligned} \quad (42)$$

(n) being the number of variables $u_1, u_2 \dots$. Hence also

$$\begin{aligned} & f \{ e^{u_1 \sqrt{-1}}, e^{u_2 \sqrt{-1}} \dots \} \\ & = \Sigma_{-\infty}^{\infty} \Sigma_{-\infty}^{\infty} \dots \left\{ \cos(r_1 m_1 + \dots) \cos(r_1 \chi \lambda_1 + \dots) \Pi \left\{ \frac{2\sqrt{-1} \chi' \lambda \{ 1 - \frac{1}{2} e(\lambda + \lambda^{-1}) \}}{\sqrt{1-e^2} - \frac{1}{2} e(\lambda - \lambda^{-1})} \right\} f(\lambda_1, \lambda_2 \dots) \right. \\ & \quad \left. + \sin(r_1 m_1 + \dots) \sin(r_1 \chi \lambda_1 + \dots) \Pi \left\{ \frac{2\sqrt{-1} \chi' \lambda \{ 1 - \frac{1}{2} e(\lambda + \lambda^{-1}) \}}{\sqrt{1-e^2} - \frac{1}{2} e(\lambda - \lambda^{-1})} \right\} f(\lambda_1, \lambda_2 \dots) \right\}. \end{aligned} \quad (43)$$

By choosing for $f\{e^{u_1\sqrt{-1}}, e^{u_2\sqrt{-1}} \dots\}$, functions expansible without sines, or without cosines, a variety of formulae may be obtained: we may instance

$$\frac{(\lambda - \lambda^{-1})\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}A_r}{\sqrt{1 - e^2} - \frac{1}{2}e(\lambda - \lambda^{-1})} = 0, \quad (44)$$

A_r having the same meaning as before.

Also,

$$\frac{\{\frac{1}{2}(\lambda + \lambda^{-1}) - e\}\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}A'_r}{\sqrt{1 - e^2} - \frac{1}{2}e(\lambda - \lambda^{-1})} = 0, \quad (45)$$

where

$$A'_r = \lambda^r e^{-\frac{re}{2}(\lambda - \lambda^{-1})} - \lambda^{-r} e^{\frac{re}{2}(\lambda - \lambda^{-1})}. \quad (46)$$

Again,

$$\frac{\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}(\lambda - \lambda^{-1})A'_r}{1 - \frac{1}{2}\frac{e}{\sqrt{1 - e^2}}(\lambda - \lambda^{-1})} + \frac{2}{r}A_r = 0, \quad (47)$$

and

$$\frac{\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}\{(\lambda + \lambda^{-1}) - \frac{1}{2}e\}A_r}{1 - \frac{1}{2}\frac{e}{\sqrt{1 - e^2}}(\lambda - \lambda^{-1})} = \{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}\{(\lambda + \lambda^{-1}) - \frac{1}{2}e\}A_r, \quad (48)$$

or, what is the same thing,

$$\frac{(\lambda - \lambda^{-1})\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}\{(\lambda + \lambda^{-1}) - \frac{1}{2}e\}A_r}{1 - \frac{1}{2}\frac{e}{\sqrt{1 - e^2}}(\lambda - \lambda^{-1})} = 0; \quad (49)$$

or, comparing with (44),

$$\frac{(\lambda^2 - \lambda^{-2})\{1 - \frac{1}{2}e(\lambda + \lambda^{-1})\}A_r}{1 - \frac{1}{2}\frac{e}{\sqrt{1 - e^2}}(\lambda - \lambda^{-1})} = 0, \quad (50)$$

which are all obtained by applying the formula (43) to the expansion of $\frac{\sin}{\cos}(\theta - \varpi)$, and comparing with the equations (25), (33).

5.

ON THE INTERSECTION OF CURVES.

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THE following theorem is quoted in a note of Chasles' *Aperçu Historique &c.*, *Mémoires de Bruxelles*, tom. XI. p. 149, where M. Chasles employs it in the demonstration of Pascal's theorem: "If a curve of the third order pass through eight of the points of intersection of two curves of the third order, it passes through the ninth point of intersection." The application in question is so elegant, that it deserves to be generally known. Consider a hexagon inscribed in a conic section. The aggregate of three alternate sides may be looked upon as forming a curve of the third order, and that of the remaining sides, a second curve of the same order. These two intersect in nine points, viz. the six angular points of the hexagon, and the three points which are the intersections of pairs of opposite sides. Suppose a curve of the third order passing through eight of these points, viz. the aggregate of the conic section passing through the angular points of the hexagon, and of the line joining two of the three intersections of pairs of opposite sides. This passes through the ninth point, by the theorem of Chasles, i.e. the three intersections of pairs of opposite sides lie in the same straight line, (since obviously the third intersection does not lie in the conic section); which is Pascal's theorem.

The demonstration of the above property of curves of the third order is one of extreme simplicity. Let $U = 0$, $V = 0$, be the equations of two curves of the third order, the curve of the same order which passes through eight of their points of intersection (which may be considered as eight perfectly arbitrary points), and a ninth arbitrary point, will be perfectly determinate. Let U_0 , V_0 , be the values of U , V , when the coordinates of this last point are written in place of x, y . Then $UV_0 - U_0V = 0$, satisfies the above conditions, or it is the equation to the curve required; but it is an equation which is satisfied by all the nine points of intersection of the two curves, i.e. any curve that passes through eight of these points of intersection, passes also through the ninth.

Consider generally two curves, $U_m = 0$, $V_n = 0$, of the orders m and n respectively, and a curve of the r^{th} order (r not less than m or n) passing through the mn points of intersection. The equation to such a curve will be of the form

$$U = u_{r-m}U_m + v_{r-n}V_n = 0,$$

u_{r-m} , v_{r-n} , denoting two polynomials of the orders $r - m$, $r - n$, with all their coefficients complete. It would at first sight appear that the curve $U = 0$ might be made to pass through as many as $\{1 + 2 \dots + (r - m + 1)\} + \{1 + 2 \dots + (r - n + 1)\} - 1$, arbitrary points, i.e.

$$\frac{1}{2}(r - m + 1)(r - m + 2) + \frac{1}{2}(r - n + 1)(r - n + 2) - 1;$$

or, what is the same thing,

$$\frac{1}{2}r(r + 3) - mn + \frac{1}{2}(r - m - n + 1)(r - m - n + 2)$$

arbitrary points, such being apparently the number of disposable constants. This is in fact the case as long as r is not greater than $m + n - 1$; but when r exceeds this, there arise, between the polynomials which multiply the disposable coefficients, certain linear relations which cause them to group themselves into a smaller number of disposable quantities. Thus, if r be not less than $m + n$, forming different polynomials of the form $x^\alpha y^\beta V_n$ [$\alpha + \beta =$ or $< n$], and multiplying by the coefficients of $x^\alpha y^\beta$ in U_m and adding, we obtain a sum $U_m V_n$, which might have been obtained by taking the different polynomials of the form $x^\gamma y^\delta U_m$ [$\gamma + \delta =$ or $< m$], multiplying by the coefficients of $x^\gamma y^\delta$ in V_n , and adding; or we have a linear relation between the different polynomials of the forms $x^\alpha y^\beta V_n$, and $x^\gamma y^\delta U_m$. In the case where r is not less than $m + n + 1$, there are two more such relations, viz. those obtained in the same way from the different polynomials $x^\alpha y^\beta \cdot x V_n$, $x^\gamma y^\delta \cdot x U_m$, and $x^\alpha y^\beta \cdot y V_n$, $x^\gamma y^\delta \cdot y U_m$, &c.; and in general, whatever be the excess of r above $m + n - 1$, the number of these linear relations is

$$1 + 2 \dots (r - m - n + 1) = \frac{1}{2}(r - m - n + 1)(r - m - n + 2).$$

Hence, if r be not less than $m + n$, the number of points through which a curve of the r^{th} order may be made to pass, in addition to the mn points which are the intersections of $U_m = 0$, $V_n = 0$, is simply $\frac{1}{2}r(r + 3) - mn$. In the case of $r = m + n - 1$, or $r = m + n - 2$, the two formulae coincide. Hence we may enunciate the theorem

“A curve of the r^{th} order, passing through the mn points of intersection of two curves of the m^{th} and n^{th} orders respectively, may be made to pass through

$$\frac{1}{2}r(r + 3) - mn + \frac{1}{2}(m + n - r - 1)(m + n - r - 2)$$

arbitrary points, if r be not greater than $m + n - 3$: if r be greater than this value, it may be made to pass through $\frac{1}{2}r(r + 3) - mn$ points only.”

Suppose r not greater than $m + n - 3$, and a curve of the r^{th} order made to pass through

$$\frac{1}{2}r(r + 3) - mn + \frac{1}{2}(m + n - r - 1)(m + n - r - 2)$$

arbitrary points, and

$$mn - \frac{1}{2}(m + n - r - 1)(m + n - r - 2)$$

of the mn points of intersection above. Such a curve passes through $\frac{1}{2}r(r+3)$ given points, and though the $mn - \frac{1}{2}(m+n-r-1)(m+n-r-2)$ latter points are not perfectly arbitrary, there appears to be no reason why the relation between the positions of these points should be such as to prevent the curve from being completely determined by these conditions. But if it be so, then the curve must pass through the remaining $\frac{1}{2}(m+n-r-1)(m+n-r-2)$ points of intersection, or we have the theorem

“If a curve of the r^{th} order (r not less than m or n , not greater than $m+n-3$) pass through

$$mn - \frac{1}{2}(m+n-r-1)(m+n-r-2)$$

of the points of intersection of two curves of the m^{th} and n^{th} orders respectively, it passes through the remaining

$$\frac{1}{2}(m+n-r-1)(m+n-r-2)$$

points of intersection.”

6.

ON THE MOTION OF ROTATION OF A SOLID BODY.

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In the fifth volume of Liouville's Journal, in a paper "Des lois géométriques qui régissent les déplacements d'un système solide," M. Olinde Rodrigues has given some very elegant formulae for determining the position of two sets of rectangular axes with respect to each other, employing rational functions of three quantities only. The principal object of the present paper is to apply these to the problem of the rotation of a solid body; but I shall first demonstrate the formulae in question, and some others connected with the same subject which may be useful on other occasions.

Let $Ax, Ay, Az; Ax_1, Ay_1, Az_1$, be any two sets of rectangular axes passing through the point A : x, y, z, x_1, y_1, z_1 , being taken for the points where these lines intersect the spherical surface described round the centre A with radius unity. Join xx_1, yy_1, zz_1 , by arcs of great circles, and through the central points of these describe great circles cutting them at right angles: these are easily seen to intersect in a certain point P . Let $Px = f, Py = g, Pz = h$; then also $Px_1 = f, Py_1 = g, Pz_1 = h$: and $\angle xPx_1 = \angle yPy_1 = \angle zPz_1 = \theta$ suppose, θ being measured from xP towards yP , yP towards zP , or zP towards xP . The cosines of f, g, h , are of course connected by the equation

$$\cos^2 f + \cos^2 g + \cos^2 h = 1.$$

Let $\alpha, \beta, \gamma; \alpha', \beta', \gamma'; \alpha'', \beta'', \gamma''$, represent the cosines of $x_1x, y_1x, z_1x; x_1y, y_1y, z_1y; x_1z, y_1z, z_1z$: these quantities are to be determined as functions of f, g, h, θ .

Suppose for a moment,

$$\angle yPz = x, \quad \angle zPx = y, \quad \angle xPy = z;$$

[Paper 6 continues on p. 29; see the neighboring pages 051–075 slice.]